Being a left adjoint is a mere proposition

The Univalent Foundations Program

hott-book:categories.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'ct:adjprop' (source lines 504-506, Lemma 9.3.2); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 9, section 3, theorem offset 1) make the counted lemma print as Lemma 9.3.2, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. Three external references are the results of other sections that the proof invokes by label; all three are navigation pointers and none is load-bearing in place of the counted proof, which is retained in full.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: being a left adjoint is a mere proposition (Lemma 9.3.2).

Allowed standard prerequisites (editorial).

- Adjunction data. A left adjoint F comes with a right adjoint G, unit eta and counit epsilon satisfying the triangle identities.. Two such quadruples are compared in the proof.
- Natural isomorphism and identity of functors. A natural isomorphism between functors in a category of functors gives an identity of the functors.. It is the step at source line 521.
- Hom-sets are sets. The hom-types of a precategory are sets.. It is why the triangle identities transport automatically at source line 533.

Source reference. Original lemma 9.2.8: sources/hott-categories/book-labelnumbers-first-edition.tex, source lines 1117–1117. Original result cited by the retained text but located in another section of the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot. It is kept as a pointer rather than a context span because retaining a result from another section as context would print it with this section's number.

Source reference. Original theorem 9.2.5: sources/hott-categories/book-labelnumbers-first-edition.tex, source lines 1103–1103. Original result cited by the retained text but located in another section of the chapter. It is a navigation pointer

only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot. It is kept as a pointer rather than a context span because retaining a result from another section as context would print it with this section's number.

Source reference. Original lemma 9.1.9: sources/hott-categories/book-labelnumbers-first-edition.tex, source lines 1115–1115. Original result cited by the retained text but located in another section of the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot. It is kept as a pointer rather than a context span because retaining a result from another section as context would print it with this section's number.

Lemma 9.3.2. *If A is a category (but B may be only a precategory), then the type "F is a left adjoint" is a mere proposition.*

Proof. Suppose we are given (G, η, ϵ) with the triangle identities and also (G', η', ϵ'). Define $\gamma : G \to G'$ to be $(G'\epsilon)(\eta' G)$, and $\delta : G' \to G$ to be $(G\epsilon')(\eta G')$. Then

$$\begin{aligned}
\delta\gamma &= (G\epsilon')(\eta G')(G'\epsilon)(\eta' G) \\
&= (G\epsilon')(GFG'\epsilon)(\eta G'FG)(\eta' G) \\
&= (G\epsilon)(G\epsilon' FG)(GF\eta' G)(\eta G) \\
&= (G\epsilon)(\eta G) \\
&= 1_G
\end{aligned}$$

using original lemma 9.2.8 and the triangle identities. Similarly, we show $\gamma\delta = 1_{G'}$, so γ is a natural isomorphism $G \cong G'$. By original theorem 9.2.5, we have an identity $G = G'$.

Now we need to know that when η and ϵ are transported along this identity, they become equal to η' and ϵ'. By original lemma 9.1.9, this transport is given by composing with γ or δ as appropriate. For η, this yields

$$(G'\epsilon F)(\eta' GF)\eta = (G'\epsilon F)(G'F\eta)\eta' = \eta'$$

using original lemma 9.2.8 and the triangle identity. The case of ϵ is similar. Finally, the triangle identities transport correctly automatically, since hom-sets are sets. □